

SOFTWARE REQUIREMENT ANALYSIS

ScamShield AI

AI-Powered Scam Detection for the Indian Digital Ecosystem

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1. Purpose & Project Vision

ScamShield AI is a multilingual scam-detection engine designed to help users identify fraudulent digital messages quickly and understand why a message is risky. The project is motivated by scams commonly delivered through everyday messaging channels, including phishing links, fake KYC requests, UPI/payment fraud, lottery scams and digital-arrest style intimidation.

The first working integration is WhatsApp, but the detection core is intentionally platform-agnostic. Channel connectors can feed messages from WhatsApp, SMS, Instagram, Facebook/Messenger and other supported sources into the same analysis pipeline.

2. Core Objectives

- Detect English, Hindi, Gujarati and Hinglish/Gujlish messages without forcing users to translate them.
- Classify messages as Safe, Suspicious or Scam and provide a confidence score.
- Explain the result using the scam type, triggering phrase(s), suspicious links and manipulation tactics.
- Provide a fast, simple response that non-technical users can act on.
- Maintain scam-pattern knowledge and analytics so new examples can improve the system over time.
- Keep the detection engine reusable across multiple digital communication channels.

3. Literature review

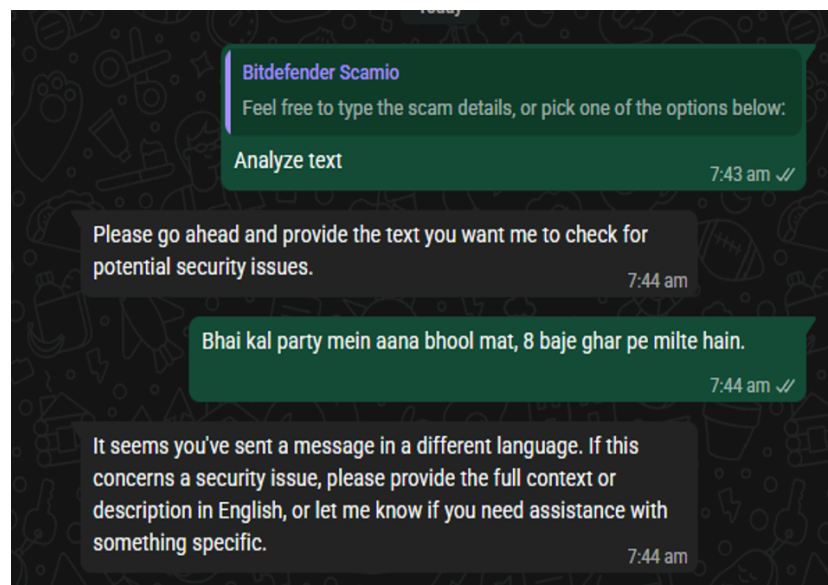
Tools	Type	Key Characteristics
Norton Genie	Standalone app (paid)	English-only; explicitly does not evaluate scams in other languages
Bitdefender Scamio	WhatsApp/web chatbot (free)	Requires account signup mid-flow; does not analyze non-English text, explicitly

		asking users to translate messages into English first
Trend Micro ScamCheck	Web tool (limited free)	Character limit issues; missed phishing scams in independent tests
McAfee Scam Detector	Bundled in security suite	Full features gated behind subscription; strong on links, weaker on plain text
Google Messages / Phone AI	On-device (Android)	Built-in, proactive, but general-purpose not India-scam-specialized
Meta AI (in WhatsApp)	General assistant	Reactive user must think to ask; not a dedicated scam-checking flow

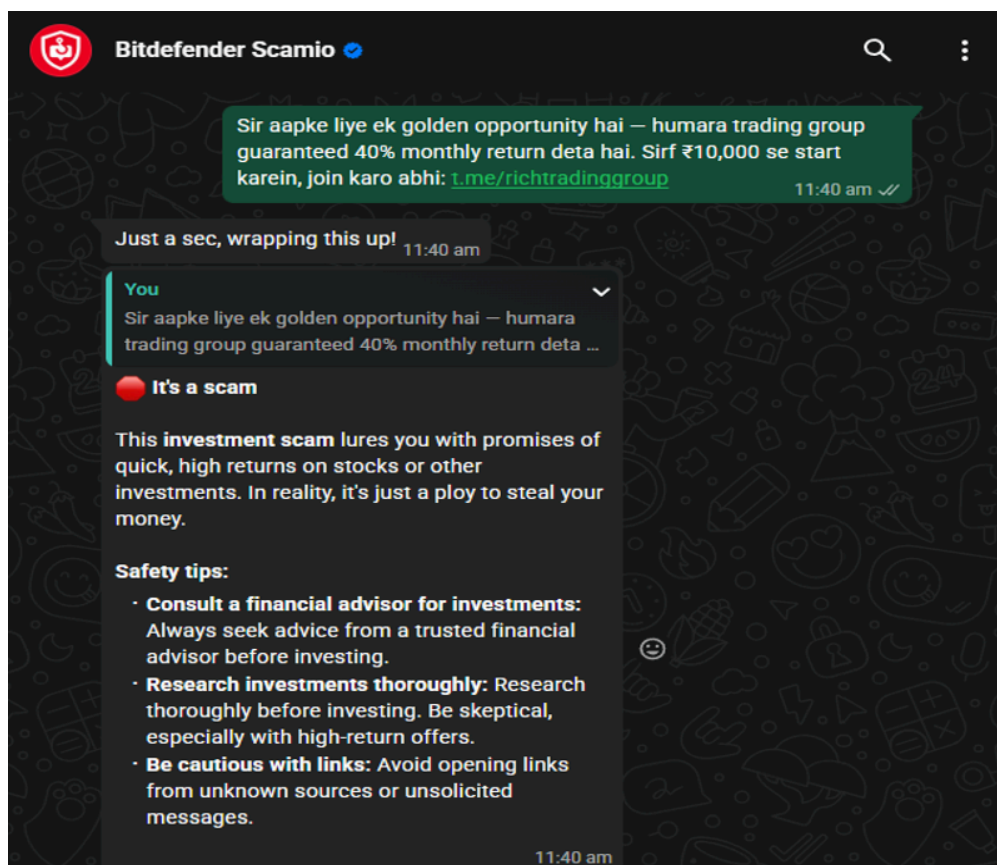
Sources: official product pages, independent reviewer testing (Safety Detectives, KSL News), and hands-on testing conducted for this project.

We didn't just read reviews, we tested Bitdefender Scamio directly with real scam-style messages. Here are our Findings:-

- Refuses to process plain conversational Hinglish:** When sent a harmless personal message ("Bhai kal party mein aana bhool mat..."), Scamio did not analyze it — it flagged the text as being in "a different language" and asked the user to translate it into English before it could help.



- **Generic, templated explanations:** Even when correctly flagged, explanations never named the specific scam type (e.g., "digital arrest"), never quoted the actual phrases used, and never flagged the isolation tactic ("don't tell anyone").



4. Scope at a Glance

In Scope	Out of Scope for Current Release
Text-message scam detection across supported channels; WhatsApp is the initial connector.	Voice-call scam detection.
Multilingual/code-mixed text analysis; URLs, urgency and sender-pattern signals.	Image/deepfake-only analysis unless text is available.
Safe / Suspicious / Scam verdict, confidence and explanation.	Telecom/network-level automatic blocking of senders.
Admin analytics, review and scam-pattern updates.	Full production-scale deployment beyond the academic evaluation environment.

5. Stakeholders & User Needs

Stakeholder	Key Need
End User	Fast, trustworthy verdicts in simple language; no technical setup.
Non-technical / vulnerable users	Clear explanation of why a message is risky and what action to take.
Administrator / Project Team	Monitor detection volume, review errors and update scam patterns.
Third-party API providers	Reliable messaging and AI services within usage limits.

6. Functional Requirements

ID	Module	Essential Requirement	Priority
1	Message Intake	Receive text from a supported channel; accept English, Hindi, Gujarati and Hinglish/Gujlish without translation.	High
2	Feature Extraction	Extract URLs, urgency phrases, sender-pattern signals and normalize code-mixed text.	High
3	Classification	Use fast ML screening plus deeper generative-AI reasoning for ambiguous/high-risk cases.	High
4	Verdict	Return Safe / Suspicious / Scam with a 0–100% confidence score.	High
5	Pattern Matching	Cross-check message/link indicators against the maintained scam-pattern database.	High
6	Explainability	Name the scam type, highlight triggering phrase(s), and identify tactics such as urgency or isolation.	High
7	User Action	Provide Report, Block Sender or Ignore actions where applicable.	Medium
8	Logging	Store the minimum required message/verdict data for analytics, review and improvement.	Medium

9	Admin Dashboard	Show scanned/flagged/safe counts, scam-type trends and model-performance information.	Medium
10	Feedback	Allow authorized review/correction of disputed classifications for model and pattern improvement.	Medium

7. Detection Pipeline (Functional View)

Every supported channel feeds the same core engine. The WhatsApp connector is the initial implementation path; other connectors can be added without redesigning the classifier.

ScamShield AI – System Workflow

Platform-agnostic core: channel connectors feed the same detection engine.

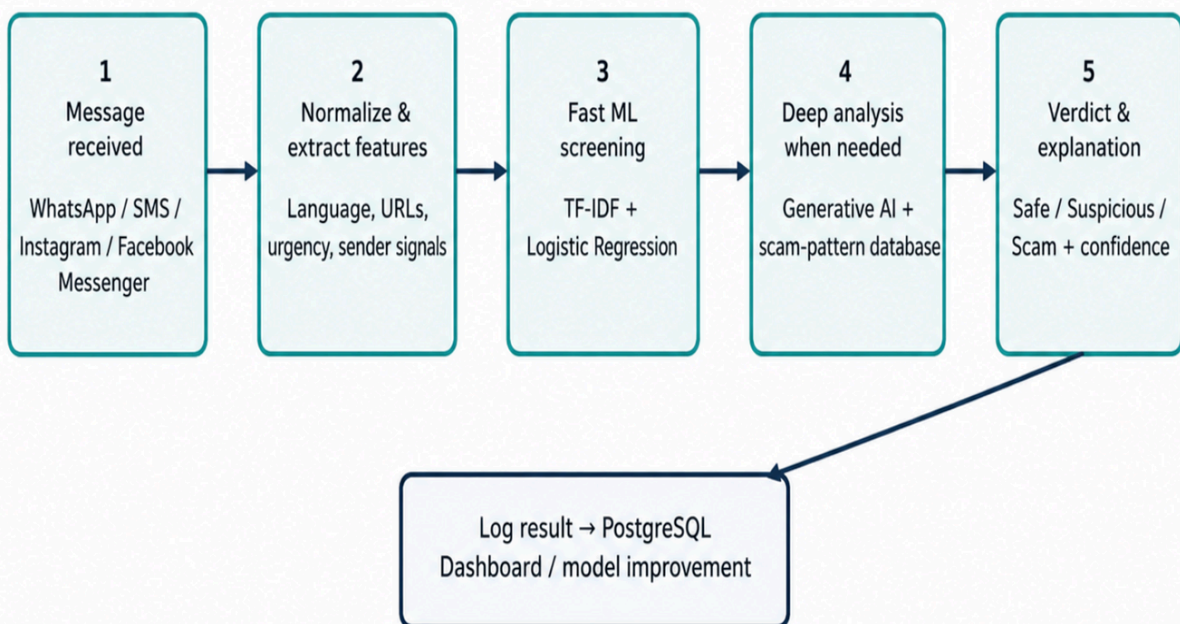


Figure 1. Simplified platform-agnostic detection workflow.

8. Processing Logic

Stage	What the system does	Output
1. Intake	Receives message from channel connector and acknowledges receipt.	Raw text + basic metadata
2. Normalize / Extract	Handles code-mixed text; extracts links, urgency queues and sender signals.	Analysis features
3. Fast Screening	TF-IDF + Logistic Regression provides low-latency statistical score.	Initial risk score
4. Deep Reasoning	Ambiguous or high-risk cases can be passed to generative AI for contextual analysis.	Contextual risk + rationale
5. Pattern Cross-check	Compares content/link indicators with known scam patterns.	Supporting evidence
6. Verdict	Combines evidence into Safe / Suspicious / Scam + confidence.	User-facing result
7. Feedback Loop	Logs result; dashboard review can identify misclassifications and update patterns.	Improvement data

9. Example User Result

Message Signal	Expected Explanation
“Pay now / blocked today”	Urgency or pressure tactic detected.
Shortened / suspicious link	Potential phishing or malicious destination indicator.
“Do not tell anyone”	Isolation/manipulation tactic detected.
Fake KYC / UPI / digital arrest language	Specific scam category named in the explanation.
Normal conversational message	Should remain Safe without an unnecessary warning.

10. Non-Functional Requirements

Category	Requirement / Target
Performance	Typical text verdict within a few seconds to preserve a real-time chat experience.
Usability	No technical expertise or separate app required for the initial WhatsApp flow; explanations must be plain-language.
Multilingual Support	English, Hindi, Gujarati and Hinglish/Gujlish, including transliterated/code-mixed text.
Explainability	Every non-safe verdict includes a human-readable reason, scam type and triggering evidence.
Security & Privacy	Use HTTPS; restrict dashboard access; minimize retention of personally identifiable information and stored content.
Reliability	Aim for high availability; if a dependency fails, return a clear fallback instead of silently failing.
Scalability	Keep the detection engine platform-agnostic and stateless where possible so new channel connectors can be added.
Maintainability	Scam-pattern data should be updateable without redeploying the core classifier.
Portability	Core detection API must be callable independently of WhatsApp.
Cost-effectiveness	Use free/low-cost services suitable for academic-scale deployment.

11. Data Requirements

- Labeled scam and non-scam messages in English, Hindi, Gujarati and Hinglish for training/evaluation.
- Structured scam-pattern database covering categories such as UPI fraud, fake KYC, digital arrest, lottery scams and phishing links.
- Minimal message log: sender identifier, timestamp, message/features, verdict and confidence for analytics.
- User-reported disputes/corrections to support supervised refinement.

- Data retention must be limited to what is necessary for scam detection, analytics and project evaluation.

12. Security, Privacy & Responsible Use

Because scam messages can contain personal or financial information, the system should minimize retained data, protect data in transit and restrict administrative access. The project is an assistance and awareness tool: a verdict should help the user make a safer decision, not be presented as an absolute guarantee of authenticity.

13. Key User Interactions

Use Case	Actor	Outcome
Scan Message	End User	Message is analyzed and returned with Safe / Suspicious / Scam, confidence and explanation.
Review Analytics	Administrator	Dashboard shows trends, scam types, flagged cases and model information.
Report Incorrect Verdict	End User / Administrator	Disputed result is logged for review and future improvement.

14. Technology & Deployment Requirements

Layer	Technology / Role
Language	Python
Classical ML	scikit-learn TF-IDF + Logistic Regression
Generative AI	Anthropic API / OpenAI API
Backend	FastAPI
Initial Messaging Connector	Twilio WhatsApp Sandbox API
Database	PostgreSQL
Dashboard	Streamlit
Hosting	Render application + PostgreSQL instance

15. Constraints, Assumptions & Dependencies

Type	Important Point
Constraint	WhatsApp Sandbox limits testing/distribution and may impose rate/session limits.
Constraint	Generative-AI availability, latency and usage cost can affect response time and budget.
Constraint	Academic timeline and two-member team limit the breadth of the first release.
Constraint	Novel scam formats may not be recognized until patterns/data are added.
Assumption	Users have internet access and can send/forward a suspicious text to the service.
Dependency	Twilio for WhatsApp connectivity; other future channels require their respective approved APIs/connectors.
Dependency	Anthropic/OpenAI for generative reasoning.
Dependency	Render, PostgreSQL and scikit-learn for hosting, storage and ML.

16. Acceptance Criteria

- A supported-channel message in English, Hindi, Gujarati or Hinglish receives a Safe / Suspicious / Scam verdict with confidence.
- Suspicious/Scam explanations identify the scam type and quote the specific triggering phrase(s).
- Plain conversational vernacular/code-mixed text is not rejected merely because it is non-English.
- Typical text analysis completes within a real-time-appropriate delay.
- Dashboard analytics accurately reflect logged results and support review/update workflows.
- The first release demonstrates the project's intended improvement in multilingual handling and explanation specificity over the tested baseline.

17. Requirement Summary

ScamShield AI should be treated as a reusable scam-detection service rather than a WhatsApp-only product. WhatsApp is the initial delivery channel, while the core requirements multilingual analysis, hybrid ML + generative reasoning, pattern matching, explainability, confidence scoring, feedback and analytics remain common across future digital messaging channels